

YTK Golden Gate Assembly

The YTK toolkit is a characterized collection of standardized genetic parts designed for modular, multi-part and hierarchical assembly of constructs for expression in *S. cerevisiae*.

Promoter, coding sequence and terminator parts are assembled into transcription units through a L1 assembly reaction. Backbone used for this reaction may either be pYTK096, pWS064 and pWS065 (pre-assembled integration vectors targeting URA3, LEU2 and HO loci respectively) or within intermediary vectors. These intermediary vectors can be used for multi-cassette assembly via L2 assembly reaction. For more detailed information, please refer to Lee et al.'s original publication.

Of note, an extension of the YTK toolkit referred to as the MYT toolkit is available. It contains a set of 10 integration vectors which can be used in place of pYTK096/pWS064/pWS065 to target a selection of neutral sites.

Transform the entirety of the reaction using standard *E. coli* heat shock transformation following each assembly to amplify plasmids.

Reagents

✓	Name	Quantity	Location/Notes
☐	DNA parts, see protocol for details and volumes	50fmol/ μ L each	
☐	BsaI restriction enzyme	0.5 μ L	
☐	BsmBI restriction enzyme	0.5 μ L for each reaction	

☐	T4 DNA Ligase Buffer (10x)	1 µl per reaction	
☐	Nuclease-free water	to 10 µl per reaction	

Equipment

✓	Name	Quantity	Location/Notes
☐	Thermocycler		
☐	PCR tube		
☐			

Instructions

L0 Assembly

1. Take 10x T4 DNA Ligase Buffer and thaw on ice. Keep BsmBI and T7 DNA Ligase in -20C until needed
2. Mix the following reagents in a PCR tube, **IN THIS ORDER**:

Reagent Tracker	Quantity
☐ Entry vector	0.5 µl
☐ Inserts / PCR product	Total 0.5 - 2µl
☐ Nuclease-free H ₂ O	to 10 µl **
☐ 10x T4 DNA Ligase Buffer	1 µl
☐ BsmBI	0.5 µl
☐ T7 DNA Ligase	0.5 µl

**calculate appropriate volume first

3. Gently mix reaction by pipetting up and down, then spin BRIEFLY to have liquid settle in the bottom of the tube
4. Set up program in thermocycler for **L0/L2**

Temperature	Time	Cycles
42°C	5 min	25X
16°C	5 min	
60°C	10 min	1X
80°C	10 min	1X
16°C	Hold	-

5. Use assembly mix directly to transform *E. coli*
6. Screen transformant colonies

L1 Assembly

7. Thaw 10x T4 DNA Ligase Buffer and thaw on ice. Keep BsaI and T7 DNA Ligase in -20°C until needed
8. Mix the following reagents in a PCR tube, **IN THIS ORDER:**

Reagent Tracker	Quantity
☐ Backbone	0.25 µl
☐ Part plasmids	0.5µl each
☐ Nuclease-free H ₂ O	to 10 µl **

☐ 10x T4 DNA Ligase Buffer	1 µl
☐ BsaI	0.5 µl
☐ T7 DNA Ligase	0.5 µl

**calculate appropriate volume first

9. Gently mix reaction by pipetting up and down, then spin BRIEFLY to have liquid settle in the bottom of the tube
10. Set up program in thermocycler for **L1**

Temperature	Time	Cycles
37°C	5 min	25X
16°C	5 min	
60°C	10 min	1X
80°C	10 min	1X
16°C	Hold	-

11. Use assembly mix directly to transform *E. coli*
12. Screen transformant colonies

L2 Assembly

13. Thaw 10x T4 DNA Ligase Buffer and thaw on ice. Keep BsmBI and T7 DNA Ligase in -20°C until needed
14. Mix the following reagents in a PCR tube, **IN THIS ORDER:**

Reagent Tracker	Quantity
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☐ Backbone	0.25 µl each
☐ LI TU cassettes	0.5 µl each
☐ Nuclease-free H ₂ O	to 10 µl **
☐ 10x T4 DNA Ligase Buffer	1 µl
☐ BsmBI	0.5 µl
☐ T7 DNA Ligase	0.5 µl

**calculate appropriate volume first

15. Gently mix reaction by pipetting up and down, then spin BRIEFLY to have liquid settle in the bottom of the tube
16. Set up program in thermocycler for **L0/L2**

Temperature	Time	Cycles
42°C	5 min	25X
16°C	5 min	
60°C	10 min	1X
80°C	10 min	1X
16°C	Hold	-

17. Use assembly mix directly to transform *E. coli*
18. Screen transformant colonies

Notes here

References

- Adapted from Dr Klemm's protocol